PROJECT REPORT

Script-Controlled ACL – Restrict Record Access Based on Field Value

ServiceNow Micro Project

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Class	3rd BCA
Project Area	ServiceNow · Access Control
Project Focus	Script-Controlled ACL
Timer	3 hrs

Description

This lab demonstrates how to implement a script-controlled Access Control List (ACL) to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the EEE branch are allowed to view records where the Branch field is set to EEE, while administrators retain full access.

1. MILESTONE – CREATION OF USERS AND ROLES

User Creation

1. Log in to the ServiceNow instance with admin access.
2. Navigate to **User Administration** → **Users**.
3. Click **New**.
4. User ID: **EEE User**
5. First name: **EEE**
6. Last name: **User**
7. Email: **eeeuser@gmail.com**
8. Save the record.

Role Creation

1. Navigate to **User Administration** → **Roles**.
2. Click **New**.
3. Name: **bb1**.
4. Enter a brief role purpose in Description.
5. Click **Submit**.
6. Create other roles: **bb2, bb3, bb4** using the same procedure.
7. Assign all the roles to the **EEE User**.

Purpose of the Roles

The custom roles are used by the ACL configuration to control which users can perform read, create, write and delete operations.

2. TABLE CREATION

Navigation

In the Application Navigator, search for **System Definition** → **Tables**.

Create the Custom Table

1. Click **New**.
2. Label: **Institution Details**.
3. Name: **u_institution_details**.
4. Extends for: **false**.
5. Click **Submit**.

Add Fields to u_institution_details

Field	Type
Student Roll Number	Auto Number
Student Name	Reference – User
Faculty Name	Reference – User
Branch	Choice (ECE, EEE, CSE)
Email	String
Phone Number	String
Description	Multi String

Save the table after adding the fields.

3. RECORD CREATION & READ ACL

Create Records

Create multiple records in the Student Records table with different Branch values: **ECE, EEE and CSE**.

Creation of Access Control List – Read

1. In the Application Navigator, search for **System Security** → **Access Control (ACL)**.
2. Elevate the **Security_admin** role.
3. Click **New** to create a new ACL.
4. Type: **record**
5. Operation: **read**
6. Name: **u_institution_details**
7. Active: **true**
8. Advanced: **true**
9. In Requires role, add **bb1**.
10. Add Data condition: **Branch is EEE**.

Script

```
(function () {
  // Allow admin users full access
  if (gs.hasRole('admin')) { return true; }

  // Allow only EEE branch users to see EEE records
  if (gs.hasRole('bb1')) { return true; }

  // Deny access for all others
  return false;
})();
```

Click **Submit** to save the ACL.

4. VERIFY READ ACL

Verification Procedure

1. Impersonate the EEE user and open the Student Records list.
2. Copy the table name **u_institution_details.list** and paste it in the All menu.
3. Impersonate Admin and open the Student Records list.

Verify

• User with **bb1** role can view only EEE branch records.
• User without the role cannot view any records.
• Admin user can view all records regardless of branch.

Access Decision Flow

User → ACL checks role → Data condition checks Branch → Script evaluates access → Allow / Deny

Important

The Read ACL is the field-value based restriction in this project. The scenario allows EEE branch users to view EEE records, while administrators retain full access.

5. CREATE ACL & WRITE ACL

Creation of Access Control List – Create

1. Click **New** to create a Create ACL.
2. Type: **record**
3. Operation: **create**
4. Name: **u_institution_details**
5. Active: **true**
6. In Requires role, add **bb2**.
7. No Data condition is required.
8. Save the ACL record.

Verify Create ACL

A user with **bb1 & bb2** roles can view only EEE branch records and gets the **New** button.

Creation of Access Control List – Write

1. Click **New** to create a Write ACL.
2. Type: **record**
3. Operation: **Write**
4. Name: **u_institution_details**
5. Active: **true**
6. In Requires role, add **bb3**.
7. No Data condition is required.
8. Save the ACL record.

6. DELETE ACL & FINAL VERIFICATION

Verify Write ACL

A user with **bb1, bb2 & bb3** roles can view and edit the EEE branch records along with the New button.

Creation of Access Control List – Delete

1. Click **New** to create a Delete ACL.
2. Type: **record**
3. Operation: **Delete**
4. Name: **u_institution_details**
5. Active: **true**
6. In Requires role, add **bb4**.
7. No Data condition is required.
8. Save the ACL record.

Verify Delete ACL

Users with **bb1, bb2, bb3 & bb4** roles can see, create and edit along with delete the EEE branch records.

Role / Operation Summary

Operation	Required Role
Read	bb1
Create	bb2
Write	bb3
Delete	bb4

7. TEAM LEAD – PROJECT SUBMISSION STEPS

Step 1: Login to SkillWallet.

Step 2: From the left-side panel, click on:
SkillWallet → **Access Resources**

Step 3: Go to **Group Projects**.

Step 4: Click on the **assigned project** for the respective team.

Step 5: The Team Lead will be able to see the following 4 tabs:
Team
Overview
Workspace
Kanban Board

Step 6: The Team Lead must move the project task cards to the **"To Be Reviewed"** tab.
■ This can be done only after submitting the **GitHub Link and Demo Link**.

Step 7: Once the above process is completed, the project progress will be reflected up to **90%**.
The remaining **10%** will be updated from their side.

OUTCOME

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation and deletion across forms, lists and Service Portal interfaces.